

Grade 7
Unit 4
Lesson 6 Practice

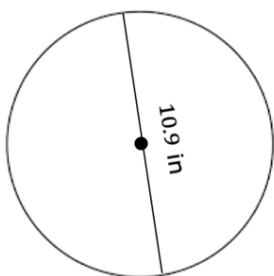
Name _____
Date _____
Period _____

1. Identify two different formulas for the circumference of a circle.

Formula 1	Formula 2
$C = \pi d$	$C = 2\pi r$

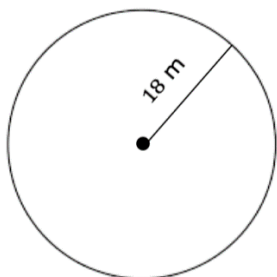
For each circle in questions 2-4, find the circumference of each circle both in terms of π and an approximate value rounded to the nearest hundredth. The circles are not drawn to scale.

2.



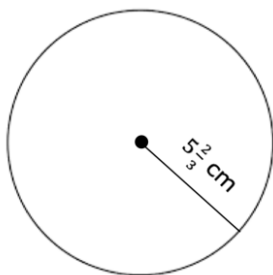
Terms of π	Approximate Value
$C = \pi d$ $= 10.9\pi$	Use $\pi = 3.14$ $C = 10.9 \cdot 3.14$ $C = 34.23$ in

3.



Terms of π	Approximate Value
$C = 2\pi r$ $C = 2 \cdot 18 \cdot \pi$ $C = 36\pi$	Use $\pi = \frac{22}{7}$ $C = 36 \cdot \frac{22}{7}$ $C = 792 \div 7$ $C = 113.14$ m

4.



Terms of π	Approximate Value
$C = 2\pi r$ $C = 2 \cdot 5\frac{2}{3} \cdot \pi$ $C = 2 \cdot \frac{17}{3} \cdot \pi$ $C = \frac{34}{3}\pi$ or $11\frac{1}{3}\pi$	Use $\pi = \frac{355}{113}$ $C = \frac{34}{3} \cdot \frac{355}{113}$ $C = \frac{12070}{339}$ or $35\frac{205}{339}$

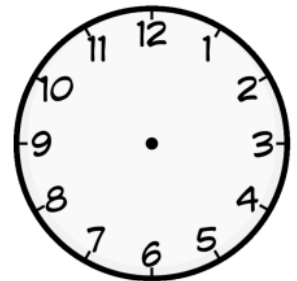
5. The Ferris wheel below has a radius of 37.25 meters. How far does a person travel in one complete rotation while riding the Ferris wheel? Use 3.14 for π and round your answer to the nearest tenth of a meter.



$$\begin{aligned} r &= 37.25 \\ C &= 2\pi r \\ C &= 2 \cdot 37.25 \cdot \pi \\ C &= 74.5 \cdot \pi \\ C &= 74.5 \cdot 3.14 \\ &\approx 233.9 \text{ m} \end{aligned}$$

6. Denise is restoring an antique clock and is painting the outer edge. If the diameter of a clock is about $9\frac{3}{5}$ inches, find the circumference. Use 3.14 for π .

$$\begin{aligned} C &= \pi d \\ C &= 9\frac{3}{5} \cdot 3.14 \\ C &= 9.6 \cdot 3.14 \\ C &= 30.144 \\ C &\approx 30.144 \text{ in} \end{aligned}$$



7. The Johnson family is getting a new fence around their circular pool. If the pool has a radius of 15.6 feet, approximately how many feet of fencing do they need to surround the pool? Use 3.14 for π .

$$\begin{aligned} C &= 2\pi r \\ C &= 2 \cdot 15.6 \cdot 3.14 \\ C &= 31.2 \cdot 3.14 \\ C &= 97.968 \\ &\approx 98 \text{ feet of fencing} \end{aligned}$$

For questions 8-10, find the circumference of each object. Round answers to the nearest tenth. Use 3.14 for π .

8.



$$d = 4\frac{1}{8} \text{ in.}$$

9.



$$r = 8.25 \text{ in.}$$

10.



$$d = 27 \text{ mm.}$$

$$C = \pi d$$

$$C = 3.14 \cdot 4.125$$

$$C = 12.9525$$

$$C \approx 13 \text{ in}$$

$$C = 2\pi r$$

$$C = 2 \cdot 8.25 \cdot 3.14$$

$$C = 16.5 \cdot 3.14$$

$$C \approx 51.81$$

$$C \approx 51.81 \text{ in}$$

$$C = \pi d$$

$$C = 3.14 \cdot 27$$

$$C = 84.78$$

$$C \approx 84.8 \text{ mm}$$